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Hollow fiber degassing module, inkjet printer, and method for degassing liquid

Abstract

A hollow fiber degassing module includes: a hollow fiber membrane bundle including a plurality of hollow fiber membranes bundled into a cylindrical shape; a housing that houses the hollow fiber membrane bundle, includes a liquid supply port and a liquid discharge port that are formed in communication with spaces between the plurality of hollow fiber membrane, and further includes a suction port formed in communication with inner spaces of the plurality of hollow fiber membranes; and an outer support having a plurality of openings formed therein and disposed between the hollow fiber membrane bundle and the housing.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a hollow fiber degassing module for degassing a liquid, an inkjet printer equipped with the hollow fiber degassing module, and a method for degassing a liquid.

BACKGROUND ART

(2) PTL 1 discloses continuous degassing of an ink using a hollow fiber degassing module that uses hollow fiber membranes and is attached to an ink channel of an inkjet printer. The hollow fiber degassing module described in PTL 1 is an external perfusion-type hollow fiber degassing module in which a hollow fiber membrane bundle including a plurality of bundled hollow fiber membranes is housed in a cylinder. An ink is supplied to the outer side of the hollow fiber membranes, and the pressure inside the spaces inside the hollow fiber membranes are reduced to degas the ink. Then the degassed ink is discharged from a discharge port formed in the side wall of the cylinder.

CITATION LIST

Patent Literature

(3) PTL 1: International Publication No. WO 2007/063720 PTL 2: Japanese Unexamined Patent Application Publication No. 2016-120489

SUMMARY OF INVENTION

Technical Problem

(4) As disclosed in PTL 2, the inventors have found that pressure drop in a hollow fiber degassing module increases abruptly when swollen hollow fiber membranes enter a hollow portion of the hollow fiber membrane bundle. On the basis of this finding, the inventors have invented a hollow fiber degassing module including a support that abuts against the inner circumferential surface of the hollow fiber membrane bundle.

(5) Then the inventors have conducted further studies and found that the pressure drop in the hollow fiber degassing module may increase abruptly not only when swollen hollow fiber membranes enter the hollow portion of the hollow fiber membrane bundle but also when the swollen hollow fiber membranes come into pressure contact with the inner circumferential surface of the housing containing the hollow fiber membrane bundle. For example, when the swollen hollow fiber membranes are in pressure contact with the inner circumferential surface of the housing, the liquid discharge port of the housing is clogged with the hollow fiber membranes, and this may cause an abrupt increase in the pressure drop in the hollow fiber degassing module.

Moreover, when the swollen hollow fiber membranes are in pressure contact with the inner circumferential surface of the housing, the gap between the hollow fiber membrane bundle and the housing is closed, and this may cause an abrupt increase in the pressure drop in the hollow fiber degassing module.

(6) Accordingly, one aspect of the present invention is to provide a hollow fiber degassing module in which an abrupt increase in pressure drop can be prevented and to provide an inkjet printer and a method for degassing a liquid.

Solution to Problem

(7) A hollow fiber degassing module according to one aspect of the present invention includes: a hollow fiber membrane bundle including a plurality of hollow fiber membranes bundled into a cylindrical shape; a housing that houses the hollow fiber membrane bundle, includes a liquid supply port and a liquid discharge port that are formed in communication with spaces between the plurality of hollow fiber membranes, and further includes a suction port formed in communication with inner spaces of the plurality of hollow fiber membranes; and an outer support having a plurality of openings formed therein and disposed between the hollow fiber membrane bundle and the housing.

(8) In this hollow fiber degassing module, since the outer support is disposed between the hollow fiber membrane bundle and the housing, the swollen hollow fiber membranes are supported by the outer support from the radially outer side of the hollow fiber membrane bundle. Therefore, even when the plurality of hollow fiber membranes swell with a liquid, the plurality of hollow fiber membranes can be prevented from coming into pressure contact with the inner circumferential surface of the housing. This configuration can prevent an abrupt increase in pressure drop in the hollow fiber degassing module due to, for example, clogging of the liquid discharge port of the housing with the hollow fiber membranes. Moreover, an abrupt increase in the pressure drop in the hollow fiber degassing module due to closing of the gap between the hollow fiber membrane bundle and the housing can be prevented. Therefore, degassing of the liquid can be performed for a long time.

(9) The outer support may have a cylindrical shape. When the outer support is cylindrical, the swollen hollow fiber membranes can be appropriately supported from the radially outer side of the hollow fiber membrane bundle.

(10) The outer support may have a mesh form. When the outer support has a mesh form, an increase in the pressure drop of the liquid passing through the outer support can be prevented while the swollen hollow fiber membranes are supported from the radially outer side of the hollow fiber membrane bundle. In this manner, an increase in the initial pressure in the hollow fiber degassing module can be prevented.

(11) The hollow fiber degassing module may further include: a first sealing portion that fixes one membrane bundle end portion of the hollow fiber membrane bundle and one support end portion of the outer support to the housing; and a second sealing portion that fixes the other membrane bundle end portion of the hollow fiber membrane bundle and the other support end portion of the outer support to the housing. In a cross section passing through one membrane bundle end portion and orthogonal to an axial direction of the housing, the first sealing portion may fill an entire region in the housing except for a hollow portion of the hollow fiber membrane bundle. In a cross section passing through the other membrane bundle end portion and orthogonal to the axial direction of the housing, the second sealing portion may fill an entire region in the housing except for the inner spaces of the plurality of hollow fiber membranes. When the first sealing portion and the second sealing portion fill the respective spaces as described above, the plurality of hollow fiber membranes divide the inner region of the housing into a first region including the spaces between the plurality of hollow fiber membranes and a second region including the inner spaces of the plurality of hollow fiber membranes. In this case, the liquid supplied from the liquid supply port can be supplied from the hollow portion of the hollow fiber membrane bundle to the spaces

between the plurality of hollow fiber membranes and then discharged from the liquid discharge port. Moreover, the inner spaces of the plurality of hollow fiber membranes can be evacuated through the suction port, and the liquid supplied from the liquid supply port can be prevented from being discharged to the suction port.

(12) The outer support may be disposed between an end surface of the first sealing portion that is on a side opposite to the second sealing portion and an end surface of the second sealing portion that is on a side opposite to the first sealing portion. When the outer support is disposed between the end surface of the first sealing portion that is on the side opposite to the second sealing portion and the end surface of the second sealing portion that is on the side opposite to the first sealing portion as described above, the hollow fiber degassing module can be produced easily, so that the cost of production of the hollow fiber degassing module can be reduced.

(13) An end portion of the outer support that is on a second sealing portion side may be embedded in the second sealing portion. When the end portion of the outer support that is on the second sealing portion side is embedded in the second sealing portion as described above, the interface between the outer support and the second sealing portion is not exposed at the end surface of the second sealing portion that is on the side opposite to the first sealing portion. In this case, even when the liquid enters the interface between the outer support and the second sealing portion, leakage of the liquid from the second sealing portion can be prevented. Moreover, when a suction pump is connected to the suction port, damage to the suction pump due to the liquid discharged from the suction port can be prevented.

(14) An end portion of the outer support that is on a first sealing portion side may be exposed from the first sealing portion. When the end portion of the outer support that is on the first sealing portion side is exposed from the first sealing portion as described above, the hollow fiber degassing module can be produced easily, so that the cost of production of the hollow fiber degassing module can be reduced.

(15) The outer support may be spaced apart from the hollow fiber membrane bundle when the plurality of hollow fiber membranes are not swollen. When the outer support is spaced apart from the hollow fiber membrane bundle when the plurality of hollow fiber membranes are not swollen as described above, the plurality of hollow fiber membranes are allowed to swell, and the swollen hollow fiber membranes can be prevented from coming into pressure contact with the inner circumferential surface of the housing. In this manner, breakage of the swollen hollow fiber membranes can be prevented.

(16) The plurality of openings of the outer support may include a first opening and a second opening adjacent to the first opening, and the outer support may have an outer communication portion that is formed between the outer support and a circumscribed circumferential surface of the outer support and allows communication between the first opening and the second opening. When the outer communication portion is formed in the outer support as described above, even if the outer support is expanded due to, for example, swelling of the plurality of hollow fiber membranes and brought into pressure contact with the inner circumferential surface of the housing, a flow channel through which the liquid flows can remain between the outer support and the inner circumferential surface of the housing. This can prevent an abrupt increase in the pressure drop in the hollow fiber degassing module.

(17) The plurality of openings of the outer support may include a first opening and a second opening adjacent to the first opening, and the outer support may have an inner communication portion that is formed between the outer support and an inscribed circumferential surface of the outer support and allows communication between the first opening and the second opening. When the inner communication portion is formed in the outer support as described above, even if the outer support is expanded due to, for example, swelling of the plurality of hollow fiber membranes and brought into pressure contact with the inner circumferential surface of the housing, a flow channel through which the liquid flows can remain between the outer support and the plurality of

hollow fiber membranes. This can prevent an abrupt increase in the pressure drop in the hollow fiber degassing module.

(18) The liquid may contain a hydrocarbon solvent. When the liquid contains the hydrocarbon solvent as described above, the plurality of hollow fiber membranes easily swell. However, since the outer support is provided, an abrupt increase in the pressure drop in the hollow fiber degassing module can be prevented.

(19) The liquid may be at least one selected from the group consisting of glycols, glycol monoalkyl ethers, glycol dialkyl ethers, glycol monoacetates, glycol diacetates, alcohols, ketones, acetates, lactates, saturated hydrocarbons, unsaturated hydrocarbons, cyclic saturated hydrocarbons, cyclic unsaturated hydrocarbons, aromatic hydrocarbons, terpenes, ethers, cyclic imides, 3-alkyl-2-oxazolidinones, N-alkyl pyrrolidones, lactone, and nitrogen-containing solvents. When the liquid is any of those described above, the plurality of hollow fiber membranes easily swell. However, since the outer support is provided, an abrupt increase in the pressure drop in the hollow fiber degassing module can be prevented.

(20) The liquid may be an UV ink or a ceramic ink. When the liquid is an UV ink or a ceramic ink as described above, the plurality of hollow fiber membranes easily swell. However, since the outer support is provided, an abrupt increase in the pressure drop in the hollow fiber degassing module can be prevented.

(21) An inkjet printer according to another aspect of the present invention is an inkjet printer in which an ink stored in an ink storage unit is supplied to an inkjet head through an ink channel. The inkjet printer includes any of the above-described hollow fiber degassing modules, and the hollow fiber degassing module is attached to the ink channel. In this inkjet printer, since any of the above-described hollow fiber degassing modules is attached to the ink channel, an abrupt increase in the pressure drop in the ink channel due to swelling of the hollow fiber membranes with the ink can be prevented. Therefore, degassing of the ink can be performed for a long time.

(22) A method for degassing a liquid according to another aspect of the present invention uses any of the above-described hollow fiber degassing modules. The method includes: supplying a liquid to the liquid supply port to thereby supply the liquid to the spaces between the plurality of hollow fiber membranes; sucking air through the suction port to degas the liquid supplied to the spaces between the plurality of hollow fiber membranes; and discharging the liquid degassed in the spaces between the plurality of hollow fiber membranes from the liquid discharge port. In this liquid degassing method, the liquid is degassed using any of the above-described hollow fiber degassing modules. Therefore, an abrupt increase in the pressure drop in the hollow fiber degassing module due to swelling of the hollow fiber membranes with the liquid can be prevented, so that degassing of the liquid can be performed for a long time.

Advantageous Effects of Invention

(23) According to the present invention, an abrupt increase in the pressure drop can be prevented.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1A schematic illustration of an inkjet printer according to an embodiment.
- (2) FIG. 2A schematic cross-sectional view of a hollow fiber degassing module according to the embodiment.
- (3) FIG. 3A partial schematic illustration of a hollow fiber membrane bundle shown in FIG. 2.
- (4) FIG. 3B partial schematic illustration of a hollow fiber membrane bundle shown in FIG. 2.
- (5) FIG. 4A cross-sectional view along line IV-IV shown in FIG. 2.
- (6) FIG. 5A perspective view of an example of an outer support.
- (7) FIG. 6A perspective view of an example of the outer support.

- (8) FIG. 7A partial enlarged view of an example of the outer support.
- (9) FIG. 8A cross-sectional view along line VIII-VIII shown in FIG. 7.
- (10) FIG. 9A cross-sectional view along line IX-IX shown in FIG. 7.
- (11) FIG. 10A schematic illustration showing the state of hollow fiber membranes in the embodiment.
- (12) FIG. 11A schematic illustration showing the state of hollow fiber membranes in a Comparative Example.
- (13) FIG. 12A schematic diagram of a test circuit.
- (14) FIG. 13A graph showing a pressure drop in Example 1 and a pressure drop in Comparative Example 1.
- (15) FIG. 14A graph showing the relation between elapsed time and the pressure drop in Example 1.

DESCRIPTION OF EMBODIMENTS

(16) A hollow fiber degassing module, an inkjet printer, and a method for degassing a liquid in an embodiment will be described in detail with reference to the drawings. In the hollow fiber degassing module in the embodiment, the hollow fiber degassing module of the present invention is applied to a hollow fiber degassing module for degassing an ink. In the drawings, the same or corresponding parts are denoted by the same symbols, and redundant description will be omitted.

(17) FIG. 1 is a schematic diagram of the inkjet printer according to the embodiment. As shown in FIG. 1, the inkjet printer **101** according to the embodiment includes mainly: an ink storage unit **102** such as an ink tank for storing an ink; an inkjet head **103** that sprays fine droplets of the ink directly to a printing medium; a first ink supply pipe **104** to which the ink is supplied from the ink storage unit **102**; a second ink supply pipe **105** for supplying the ink to the inkjet head **103**; the hollow fiber degassing module **1** according to the embodiment that is attached to the first ink supply pipe **104** and the second ink supply pipe **105** to degas the ink; a suction pump **106** for evacuation; and a suction pipe **107** that connects the suction pump **106** to the hollow fiber degassing module **1**. The first ink supply pipe **104** and the second ink supply pipe **105** are ink channels extending from the ink storage unit **102** to the inkjet head **103**. No particular limitation is imposed on the ink used for the inkjet printer **101**. Examples of the ink include water-based inks, UV inks, solvent inks, and ceramic inks.

(18) FIG. 2 is a schematic cross-sectional view of the hollow fiber degassing module according to the embodiment. FIGS. 3A and 3B are partial enlarged views of the hollow fiber membrane bundle shown in FIG. 2. FIG. 4 is a cross-sectional view along line IV-IV shown in FIG. 2. As shown in FIGS. 1 to 4, the hollow fiber degassing module **1** includes: a hollow fiber membrane bundle **3** including a plurality of hollow fiber membranes **2** bundled into a cylindrical shape; a cylindrical housing **4** that houses the hollow fiber membrane bundle **3**; an inner support **10** in contact with the inner circumferential surface **3a** of the hollow fiber membrane bundle **3**; and an outer support **20** disposed between the hollow fiber membrane bundle **3** and the housing **4**. In the hollow fiber degassing module **1**, the plurality of hollow fiber membranes **2** divide the housing **4** into a first region including spaces between the plurality of hollow fiber membranes **2** and a second region including the inner spaces of the plurality of hollow fiber membranes **2**. The first region is a region to which the ink is to be supplied, and the second region is a region to be evacuated. In the hollow fiber degassing module **1**, the ink is supplied from a hollow portion **3b** of the hollow fiber membrane bundle **3** to the spaces between the plurality of hollow fiber membranes **2** (the first region), and the inner regions of the plurality of hollow fiber membranes **2** (the second region) are evacuated to degas the ink. The hollow portion **3b** is a hollow portion located in a radially central portion of the hollow fiber membrane bundle **3**.

(19) The hollow fiber membranes **2** are membranes of hollow fibers that are permeable to gases but do not allow liquids to pass therethrough. The hollow fiber membranes **2** have the property of swelling with a liquid such as an ink. No particular limitation is imposed on the material of the

hollow fiber membranes **2**, their membrane shape, their membrane form, etc. Examples of the material of the hollow fiber membranes **2** include: polyolefin-based resins such as polypropylene, polyethylene, and polymethylpentene; silicon-based resins such as polydimethylsiloxane and its copolymers; and fluororesins such as PTFE and vinylidene fluoride. Examples of the membrane shape (the shape of the side walls) of the hollow fiber membranes **2** include porous membranes, microporous membranes, and homogeneous membranes with no pores (non-porous membranes). Examples of the membrane form of the hollow fiber membranes **2** include: symmetric membranes (homogeneous membranes) whose overall chemical or physical structure is homogeneous; and asymmetric membranes (heterogeneous membranes) whose chemical or physical structure varies from place to place. The asymmetric membrane (heterogeneous membrane) is a membrane including a nonporous dense layer and a porous portion. In this case, the dense layer may be formed in any portion in the membrane such as a surface layer portion of the porous membrane, its inner portion, etc. The term heterogeneous membrane encompasses composite membranes including different chemical structures and multilayer membranes such as three-layer structure membranes. In particular, a heterogeneous membrane using a poly 4-methylpentene-1 resin has a dense layer that blocks liquids and is therefore particularly preferable for degassing liquids other than water such as inks. For hollow fibers used for external perfusion-type modules, it is preferable that a dense layer is formed on the outermost surface of the hollow fibers.

(20) The hollow fiber membrane bundle **3** can be formed, for example, from a hollow fiber membrane sheet (not shown) including a plurality of hollow fiber membranes **2** woven into a bamboo blind form. In this case, for example, the hollow fiber membrane sheet used to form the hollow fiber membrane bundle **3** may include 10 to 40 hollow fiber membranes **2** per inch. The hollow portion **3b** serving as an ink channel is formed on the radially inner side of the hollow fiber membrane bundle **3**. The hollow portion **3b** is defined by the inner circumferential surface **3a** of the hollow fiber membrane bundle **3**.

(21) The housing **4** includes a casing **5**, a first lid portion **6**, and a second lid portion **7**.

(22) The casing **5** is a member that houses the hollow fiber membrane bundle **3**. The casing **5** is formed into a cylindrical shape extending in an axial direction **L**, and both end portions of the casing **5** are open. The first lid portion **6** is attached to one opening end portion **5a** that is one opening end portion of the casing **5**, and the second lid portion **7** is attached to the other opening end portion **5b** that is the other opening end portion of the casing **5**. The first lid portion **6** and the second lid portion **7** are attached to the casing **5** by, for example, screwing, fitting, adhering, etc.

(23) The first lid portion **6** is formed into a tapered shape whose diameter decreases as the distance from the casing **5** increases. A liquid supply port **6a** is formed at an end portion of the first lid portion **6**. An inner space in communication with the liquid supply port **6a** is formed inside the first lid portion **6**. The liquid supply port **6a** is an opening formed in the first lid portion **6** to supply the ink to the spaces between the plurality of hollow fiber membranes **2**. The liquid supply port **6a** is, for example, circular. The liquid supply port **6a** is formed on a center axis **L1** of the casing **5**. A connection portion **6b** to which the first ink supply pipe **104** is to be detachably connected extends in the axial direction **L** from the liquid supply port **6a**. The connection portion **6b** is formed into a cylindrical shape, and a female thread **6c** into which the first ink supply pipe **104** is to be screwed is formed on the inner circumferential surface of the connection portion **6b**. The connection between the connection portion **6b** and the first ink supply pipe **104** is not limited to screwing, and they may be connected by, for example, fitting.

(24) The second lid portion **7** is formed into a tapered shape whose diameter decreases as the distance from the casing **5** increases. A suction port **7a** is formed at an end portion of the second lid portion **7**. An inner space in communication with the suction port **7a** is formed inside the second lid portion **7**. The suction port **7a** is an opening formed in the second lid portion **7** to evacuate the inner spaces of the plurality of hollow fiber membranes **2**. The suction port **7a** is, for example, circular. The suction port **7a** is formed on the center axis **L1** of the casing **5**. A connection portion **7b** to

which the suction pipe **107** is to be detachably connected extends in the axial direction L from the suction port **7a**. The connection portion **7b** is formed into a cylindrical shape, and a female thread **7c** into which the suction pipe **107** is to be screwed is formed on the inner circumferential surface of the connection portion **7b**. The connection between the connection portion **7b** and the suction pipe **107** is not limited to screwing, and they may be connected by, for example, fitting.

(25) A liquid discharge port **5d** is formed in a side wall **5c** of the casing **5**. The liquid discharge port **5d** is an opening formed in the casing **5** to discharge the ink from the spaces between the plurality of hollow fiber membranes **2**. The liquid discharge port **5d** is, for example, circular. The liquid discharge port **5d** is formed in a position displaced from the center of the casing **5** in the axial direction L toward the other opening end portion **5b**. A connection portion **5e** to which the second ink supply pipe **105** is to be detachably connected extends from the liquid discharge port **5d** in a direction orthogonal to the axial direction L. The connection portion **5e** is formed into a cylindrical shape, and a female thread **5f** into which the second ink supply pipe **105** is to be screwed is formed on the inner circumferential surface of the connection portion **5e**. The connection between the liquid discharge port **5d** and the second ink supply pipe **105** is not limited to screwing, and they may be connected by, for example, fitting.

(26) From the viewpoint of ease of production, it is preferable that the casing **5**, the first lid portion **6**, and the second lid portion **7** are made of a resin. In this case, the casing **5**, the first lid portion **6**, and the second lid portion **7** can be produced by injection molding. Preferably, the casing **5**, the first lid portion **6**, and the second lid portion **7** are of a color that does not allow UV light to pass therethrough, e.g., a black color, in consideration of the case where the ink used is an UV ink.

(27) One membrane bundle end portion **3e** of the hollow fiber membrane bundle **3** is fixed to one opening end portion **5a** of the casing **5** through a first sealing portion **8**, and the other membrane bundle end portion **3f** of the hollow fiber membrane bundle **3** is fixed to the other opening end portion **5b** of the casing **5** through a second sealing portion **9**.

(28) The first sealing portion **8** is formed of a resin. Examples of the resin used for the first sealing portion **8** include epoxy resins, urethane resins, UV curable resins, and polyolefin resins such as polyethylene and polypropylene. In a cross section passing through one membrane bundle end portion **3e** of the hollow fiber membrane bundle **3** and orthogonal to the axial direction L, the first sealing portion **8** fills the entire region except for the hollow portion **3b**. Specifically, the first sealing portion **8** fills the spaces between the hollow fiber membranes **2**, the inner spaces of the plurality of hollow fiber membranes **2**, and the space between the hollow fiber membrane bundle **3** and the inner wall of the casing **5** (see FIG. 3A). The plurality of hollow fiber membranes **2** are exposed at an end surface **8b** of the first sealing portion **8** that is on the side opposite to the second sealing portion **9**. A communication port **8a** that serves as an opening of the hollow portion **3b** and allows communication between the hollow portion **3b** and the liquid supply port **6a** is formed in the first sealing portion **8**. Therefore, the ink supplied from the liquid supply port **6a** to the inner space of the first lid portion **6** is supplied to the casing **5** only from the communication port **8a**.

Specifically, the ink supplied from the liquid supply port **6a** to the housing **4** is supplied to the hollow portion **3b** of the hollow fiber membrane bundle **3** only from the communication port **8a** and then supplied from the hollow portion **3b** to the spaces between the plurality of hollow fiber membranes **2**.

(29) The second sealing portion **9** is formed of the same resin as that for the first sealing portion **8**. In a cross section passing through the other membrane bundle end portion **3f** of the hollow fiber membrane bundle **3** and orthogonal to the axial direction L, the second sealing portion **9** fills the entire region except for the inner spaces of the plurality of hollow fiber membranes **2**. Specifically, the second sealing portion **9** does not fill the inner spaces of the plurality of hollow fiber membranes **2** but fills the spaces between the plurality of hollow fiber membranes **2**, the space between the hollow fiber membrane bundle **3** and the inner wall of the casing **5**, and the hollow portion **3b** (see FIG. 3B). The plurality of hollow fiber membranes **2** are exposed at an end surface

9b of the second sealing portion **9** that is on the side opposite to the first sealing portion **8**. Communication holes **9a** that serve as openings of the inner spaces of the plurality of hollow fiber membranes **2** and allow communication between the suction port **7a** and the inner spaces of the plurality of hollow fiber membranes **2** are formed in the second sealing portion **9**. Therefore, the ink supplied from the communication port **8a** to the hollow portion **3b** does not flow to the second lid portion **7** side through the second sealing portion **9** but passes through the spaces between the plurality of hollow fiber membranes **2** and is discharged from the liquid discharge port **5d**. Since the inner spaces of the plurality of hollow fiber membranes **2** are in communication with the suction port **7a**, the inner spaces of the plurality of hollow fiber membranes **2** are reduced in pressure by sucking air through the suction port **7a** using the suction pump **106**.

(30) The first sealing portion **8** fixes one membrane bundle end portion **3e** of the hollow fiber membrane bundle **3** to the casing **5** such that, for example, a center axis **L2** of one membrane bundle end portion **3e** of the hollow fiber membrane bundle **3** and the center axis **L1** of the casing **5** are coaxial with each other. The second sealing portion **9** fixes the other membrane bundle end portion **3f** of the hollow fiber membrane bundle **3** to the casing **5** such that, for example, a center axis **L3** of the other membrane bundle end portion **3f** of the hollow fiber membrane bundle **3** and the center axis **L1** of the casing **5** are coaxial with each other. Alternatively, the second sealing portion **9** may fix the other membrane bundle end portion **3f** of the hollow fiber membrane bundle **3** to the casing **5** such that, for example, the center axis **L3** of the other membrane bundle end portion **3f** of the hollow fiber membrane bundle **3** is offset from the center axis **L1** of the casing **5** toward the side opposite to the discharge port **5d**. Alternatively, the second sealing portion **9** may fix the other membrane bundle end portion **3f** of the hollow fiber membrane bundle **3** to the casing **5** such that, for example, the hollow fiber membrane bundle **3** is inclined with respect to the center axis **L1** of the casing **5** so as to be spaced apart from the liquid discharge port **5d**.

(31) The ratio of the inner diameter of the casing **5** to the length of the hollow fiber membrane bundle **3** in the axial direction **L** may be, for example, 1:1 to 1:6.

(32) The inner support **10** is a member for supporting the plurality of hollow fiber membranes **2** (the hollow fiber membrane bundle **3**) from the radially inner side of the hollow fiber membrane bundle **3**. The inner support **10** is formed into a cylindrical shape (pipe shape). The outer diameter of the inner support **10** is approximately the same as the inner diameter of the hollow fiber membrane bundle **3**. The thickness of the inner support **10** may be appropriately set so long as, for example, the swollen hollow fiber membranes **2** can be supported. The thickness of the inner support **10** may be, for example, from 0.5 mm to 5 mm inclusive. A plurality of openings **11** are formed in the inner support **10**. The inner support **10** is formed into, for example, a mesh form. When the inner support **10** is formed into a mesh form, the meshes correspond to the plurality of openings **11**.

(33) One support end portion **10c**, which is one end portion of the inner support **10** in the axial direction **L**, is fixed to the casing **5** (the housing **4**) through the first sealing portion **8**, and the other support end portion **10d**, which is the other end portion of the inner support **10** in the axial direction **L**, is fixed to the casing **5** (the housing **4**) through the second sealing portion **9**. The first sealing portion **8** and the second sealing portion **9** also fill the space between the hollow fiber membrane bundle **3** and the inner support **10** and some of the openings **11** of the inner support **10**.

(34) The inner support **10** is disposed between the end surface **8b** of the first sealing portion **8** and the end surface **9b** of the second sealing portion **9**. Specifically, the inner support **10** does not protrude from the end surface **8b** of the first sealing portion **8** and from the end surface **9b** of the second sealing portion **9**. The inner support **10** may have the same length as the hollow fiber membrane bundle **3** in the axial direction **L** or may be shorter than the hollow fiber membrane bundle **3**. Among both ends of the inner support **10** in the axial direction **L**, the end on the first sealing portion **8** side is defined as an end **10a**, and the end on the second sealing portion **9** side is defined as an end **10b**. In this case, the end **10a** of the inner support **10** may be exposed at the end

surface **8b** of the first sealing portion **8**, and the end **10b** may be embedded in the second sealing portion **9**. Alternatively, the end **10a** of the inner support **10** may be embedded in the first sealing portion **8**, and the end **10b** may be exposed at the end surface **9b** of the second sealing portion **9**. Alternatively, the end **10a** of the inner support **10** may be embedded in the first sealing portion **8**, and the end **10b** may be embedded in the second sealing portion **9**. In the example shown in FIGS. **2** and **3**, the inner support **10** is shorter than the hollow fiber membrane bundle **3** in the axial direction **L**. In this case, the end **10a** of the inner support **10** is exposed at the end surface **8b** of the first sealing portion **8**, and the end **10b** of the inner support **10** is embedded in the second sealing portion **9**.

(35) FIGS. **5** and **6** are perspective views of examples of the outer support. As shown in FIGS. **2** to **6**, the outer support **20** is a member for supporting the swollen hollow fiber membranes **2** (the hollow fiber membrane bundle **3**) from the radially outer side of the hollow fiber membrane bundle **3**. The outer support **20** is formed into a cylindrical shape (pipe shape). The outer diameter of the outer support **20** is equal to or larger than the outer diameter of the hollow fiber membrane bundle **3** and equal to or smaller than the inner diameter of the casing **5**. The outer support **20** may abut against an outer circumferential surface **3g** of the hollow fiber member bundle **3**, may abut against an inner circumferential surface **5g** of the casing **5**, or may be spaced apart from the outer circumferential surface **3g** of the hollow fiber member bundle **3** and from the inner circumferential surface **5g** of the casing **5**. In the example shown in the figures, the outer support **20** is spaced apart from the outer circumferential surface **3g** of the hollow fiber membrane bundle **3** and from the inner circumferential surface **5g** of the casing **5**. The thickness of the outer support **20** may be appropriately set so long as, for example, the swollen hollow fiber membranes **2** can be supported. The thickness of the outer support **20** may be, for example, from 0.5 mm to 5 mm inclusive.

(36) A plurality of openings **21** are formed in the outer support **20**. The outer support **20** is formed, for example, into a mesh form. When the outer support **20** is formed into a mesh form, the meshes correspond to the plurality of openings **21**. The mesh form is a shape in which a plurality of linear portions **22** extending in different directions are connected such that the plurality of linear portions **22** form the plurality of openings **21**. Examples of the shape of the openings **21** include square shapes, rectangular shapes, pentagonal shapes, hexagonal shapes, circular shapes, and elliptical shapes. The aperture ratio of the outer support **20** may be, for example, in the range of 10% or more, preferably in the range of from 20% to 80% inclusive, and still more preferably in the range of from 30% to 60% inclusive. The aperture ratio of the outer support **20** is the ratio of the total opening area of the openings **21** to the projected area of the outer support **20** including all the openings **21** when the outer support **20** is cut and flattened into a flat sheet shape. When both the inner support **10** and the outer support **20** are formed into mesh forms, the mesh form of the inner support **10** and the mesh form of the outer support **20** may be the same or different.

(37) The outer support **20** shown in FIG. **5** includes: a plurality of first linear portions **22a** extending in a direction parallel to the axial direction of the outer support **20** and arranged in a circular pattern; and a plurality of second linear portions **22b** extending in a circular shape about the axial line of the outer support **20** and connected to the first linear portions **22a**. The openings **21** of the outer support **20** shown in FIG. **5** have a square shape. The outer support **20** shown in FIG. **6** includes: a plurality of first linear portions **22a** extending in a direction inclined at a prescribed angle with respect to the axial direction of the outer support **20**; and a plurality of second linear portions **22b** extending in a direction inclined opposite to the inclination direction of the first linear portions **22a** at the prescribed angle with respect to the axial direction of the outer support **20** and connected to the first linear portions **22a**. The openings **21** of the outer support **20** shown in FIG. **6** have a rhombic shape.

(38) The linear portions **22** may have, for example, a polygonal cross-sectional shape, a circular cross-sectional shape, etc. The line diameter of the linear portions **22** may be appropriately set so long as, for example, the swollen hollow fiber membranes **2** can be supported. From the viewpoint

of ease of production, it is preferable that the outer support **20** is formed of, for example, a resin. Examples of the resin used for the outer support **20** include polypropylenes and polyethylenes, and preferred examples include ultra-high molecular weight polyethylenes and high-density polyethylenes.

(39) FIG. **7** is a partial enlarged view of an example of the outer support. FIG. **8** is a cross-sectional view along line VIII-VIII in FIG. **7**. FIG. **9** is a cross-sectional view along line IX-IX in FIG. **7**. As shown in FIGS. **7** to **9**, one of the plurality of openings **21** of the outer support **20** is defined as a first opening **21a**, and an opening **21** adjacent to the first opening **21a** is defined as a second opening **21b**. A virtual circumferential surface circumscribing the outer support **20** is defined as a circumscribed circumferential surface VF1, and a virtual circumferential surface inscribed in the outer support **20** is defined as an inscribed circumferential surface VF2.

(40) In this case, the outer support **20** has an outer communication portion **23** formed between the outer support **20** and the circumscribed circumferential surface VF1 to allow communication between the first opening **21a** and the second opening **21b**. The outer communication portion **23** is a recessed portion of the outer support **20** that is recessed from the circumscribed circumferential surface VF1. One outer communication portion **23** may be formed between each adjacent pair of openings **21** or may be formed between only some adjacent pairs of openings **21**.

(41) The outer support **20** has an inner communication portion **24** formed between the outer support **20** and the inscribed circumferential surface VF2 to allow communication between the first opening **21a** and the second opening **21b**. The inner communication portion **24** is a recessed portion of the outer support **20** that is recessed from the inscribed circumferential surface VF2. One inner communication portion **24** may be formed between each adjacent pair of openings **21** or may be formed between only some adjacent pairs of openings **21**.

(42) As shown in FIGS. **2** and **3**, one support end portion **20c**, which is one end portion of the outer support **20** in the axial direction L, is fixed to the casing **5** (the housing **4**) through the first sealing portion **8**, and the other support end portion **20d**, which is the other end portion of the outer support **20** in the axial direction L, is fixed to the casing **5** (the housing **4**) through the second sealing portion **9**. The first sealing portion **8** and the second sealing portion **9** fill also the space between the hollow fiber membrane bundle **3** and the outer support **20** and some of the openings **21** of the outer support **20**.

(43) The outer support **20** is disposed between the end surface **8b** of the first sealing portion **8** and the end surface **9b** of the second sealing portion **9**. Specifically, the outer support **20** does not protrude from the end surface **8b** of the first sealing portion **8** and from the end surface **9b** of the second sealing portion **9**. The outer support **20** may have the same length as the hollow fiber membrane bundle **3** in the axial direction L or may be shorter than the hollow fiber membrane bundle **3**. Among both ends of the outer support **20** in the axial direction L, the end on the first sealing portion **8** side is defined as an end **20a**, and the end on the second sealing portion **9** side is defined as an end **20b**. In this case, the end **20a** of the outer support **20** may be exposed at the end surface **8b** of the first sealing portion **8**, and the end **20b** may be embedded in the second sealing portion **9**. Alternatively, the end **20a** of the outer support **20** may be embedded in the first sealing portion **8**, and the end **20b** may be exposed at the end surface **9b** of the second sealing portion **9**. Alternatively, the end **20a** of the outer support **20** may be embedded in the first sealing portion **8**, and the end **20b** may be embedded in the second sealing portion **9**. In the example shown in FIGS. **2** and **3**, the outer support **20** is shorter than the hollow fiber membrane bundle **3** in the axial direction L. In this case, the end **20a** of the outer support **20** is exposed at the end surface **8b** of the first sealing portion **8**, and the end **20b** of the outer support **20** is embedded in the second sealing portion **9**.

(44) Next a method for degassing an ink using the hollow fiber degassing module **1** will be described.

(45) The ink is supplied from the ink storage unit **102** to the first ink supply pipe **104**. Then the ink

is supplied from the liquid supply port **6a** to the inner space of the first lid portion **6**. The ink supplied to the inner space of the first lid portion **6** passes through the communication port **8a** and is supplied to the hollow portion **3b**. The ink supplied to the hollow portion **3b** passes through the plurality of openings **11** of the inner support **10**, is supplied to the spaces between the plurality of hollow fiber membranes **2**, passes through these spaces, and flows toward the radially outer side of the casing **5** (the hollow fiber membrane bundle **3**). Specifically, the ink supplied to the hollow portion **3b** is supplied to the outer side of each of the plurality of hollow fiber membranes **2** within the casing **5**. In this case, the housing **4** is evacuated through the suction port **7a** using the suction pump **106**, and the inner spaces of the plurality of hollow fiber membranes **2** are thereby reduced in pressure. Then, when the ink passes through the spaces between the plurality of hollow fiber membranes **2**, gasses dissolved in the ink and bubbles therein are drawn into the inner spaces of the plurality of hollow fiber membranes **2**. The ink is thereby degassed. Then the degassed ink passes through the plurality of openings **21** of the outer support **20**, is supplied to the gap between the hollow fiber membrane bundle **3** and the inner circumferential surface **5g** of the casing **5**, passes through the gap, and is discharged from the liquid discharge port **5d** to the second ink supply pipe **105**. The ink discharged to the second ink supply pipe **105** passes through the second ink supply pipe **105** and is supplied to the inkjet head **103**.

(46) In this case, the plurality of hollow fiber membranes **2** gradually swell with the ink over time. The rate and degree of swelling of the hollow fiber membranes **2** vary depending on the material of the hollow fiber membranes **2**, the shape of the membranes, the form of the membranes, etc. and also vary depending on the type of ink. For example, when a polyolefin-based resin such as poly-4 methylpentene-1 is used as the material of the hollow fiber membranes **2** and the ink used is a ceramic ink containing a ceramic powder dispersed in a solvent such as a hydrocarbon solvent, the rate of swelling of the hollow fiber membranes **2** and the degree of swelling are particularly large. As the plurality of hollow fiber membranes **2** swell, they bulge while distorted toward the radially inner and outer sides of the hollow fiber membrane bundle **3**. Specifically, the swollen hollow fiber membranes **2** expand toward the radially inner side of the hollow fiber membrane bundle **3** and try to enter the hollow portion **3b**. Moreover, the swollen hollow fiber membranes **2** expand toward the radially outer side of the hollow fiber membrane bundle **3**, are pressed against the inner circumferential surface **5g** of the casing **5**, and try to enter the liquid discharge port **5d**.

(47) No particular limitation is imposed on the solvent used for the ceramic ink so long as the effects of the present invention are not impaired, and any known solvent may be used. Specific examples of the solvent include: glycols such as ethylene glycol, diethylene glycol, and triethylene glycol; glycol monoalkyl ethers such as 3-methoxy-3-methylbutanol and 3-methoxybutanol; glycol dialkyl ethers such as diethylene glycol diethyl ether, diethylene glycol methyl ethyl ether, diethylene glycol methyl butyl ether, triethylene glycol methyl butyl ether, and tetraethylene glycol dimethyl ether; glycol monoacetates such as ethylene glycol monobutyl ether acetate, diethylene glycol monobutyl ether acetate, and propylene glycol monomethyl ether acetate; glycol diacetates; alcohols such as ethanol, n-propanol, isopropanol, n-butanol, 2-butanol, and 2-methyl-1-propanol; ketones such as acetone, methyl ethyl ketone, methyl-n-propyl ketone, methyl isopropyl ketone, methyl-n-butyl ketone, methyl isobutyl ketone, methyl-n-amyl ketone, methyl isoamyl ketone, diethyl ketone, ethyl-n-propyl ketone, ethyl isopropyl ketone, ethyl-n-butyl ketone, ethyl isobutyl ketone, di-n-propyl ketone, diisobutyl ketone, cyclohexanone, methylcyclohexanone, and isophorone; acetates such as methyl acetate, ethyl acetate, n-propyl acetate, isopropyl acetate, n-butyl acetate, isobutyl acetate, hexyl acetate, octyl acetate, 2-methylpropyl acetate, and 3-methylbutyl acetate; lactates such as methyl lactate, ethyl lactate, and butyl lactate; saturated hydrocarbons such as n-hexane, isohexane, n-nonane, isononane, dodecane, and isododecane; unsaturated hydrocarbons such as 1-hexene, 1-heptene, and 1-octene; cyclic saturated hydrocarbons such as cyclohexane, cycloheptane, cyclooctane, cyclodecane, and decalin; cyclic unsaturated hydrocarbons such as cyclohexene, cycloheptene, cyclooctene, 1,1,3,5,7-cyclooctatetraene, and

cyclododecene; aromatic hydrocarbons such as benzene, toluene, and xylene; terpenes; ethers; cyclic imides; 3-alkyl-2-oxazolidinones such as 3-methyl-2-oxazolidinone and 3-ethyl-2-oxazolidinone; N-alkyl pyrrolidones such as N-methyl-2-pyrrolidone and N-ethyl-2-pyrrolidone; lactones such as γ -butyrolactone and ϵ -caprolactone; and nitrogen-containing solvents such as β -alkoxypropionamides.

(48) As shown in FIG. 11, in a hollow degassing module in a Comparative Example that does not include the inner support and the outer support, swollen hollow fiber membranes 2 come into pressure contact with the inner circumferential surface 5g of the casing 5. In this case, the swollen hollow fiber membranes 2 enter the liquid discharge port 5d, and the liquid discharge port 5d is clogged with the swollen hollow fiber membranes 2. Moreover, the gap between the hollow fiber membrane bundle 3 and the casing 5 is closed. Therefore, the ink channel is narrowed or closed, and the pressure drop of the ink increases abruptly.

(49) However, as shown in FIG. 10, in the hollow fiber degassing module in the present embodiment, since the outer support 20 is disposed between the hollow fiber membrane bundle 3 and the casing 5, the swollen hollow fiber membranes 2 are unlikely to come into pressure contact with the inner circumferential surface 5g of the casing 5. Therefore, the swollen hollow fiber membranes 2 are prevented from entering the liquid discharge port 5d, and the liquid discharge port 5d is prevented from being clogged. Moreover, the gap between the hollow fiber membrane bundle 3 and the casing 5 is prevented from being closed. Since the ink channel is prevented from being narrowed or clogged, an abrupt increase in the pressure drop of the ink can be prevented.

(50) Since the inner support 10 is in contact with the inner circumferential surface 3a of the hollow fiber membrane bundle 3 that defines the hollow portion 3b, the swollen hollow fiber membranes 2 are prevented from entering the hollow portion 3b. Therefore, in the hollow fiber degassing module 1, even when the plurality of hollow fiber membranes 2 swell, the ink channel is prevented from being narrowed or clogged, so that an abrupt increase in the pressure drop of the ink can be prevented.

(51) As described above, since the hollow fiber degassing module 1 according to the present embodiment includes the outer support 20 disposed between the hollow fiber membrane bundle 3 and the casing 5, the swollen hollow fiber membranes 2 are supported by the outer support 20 from the radially outer side of the hollow fiber membrane bundle 3. Therefore, even when the plurality of hollow fiber membranes 2 swell with the ink, the plurality of hollow fiber membranes 2 are prevented from coming into pressure contact with the inner circumferential surface 5g of the casing 5. This can prevent an abrupt increase in the pressure drop in the hollow fiber degassing module 1 due to clogging of the liquid discharge port 5d of the casing 5 with the plurality of hollow fiber membranes 2. Moreover, an abrupt increase in the pressure drop in the hollow fiber degassing module 1 due to closing of the gap between the hollow fiber membrane bundle 3 and the casing 5 can be prevented. Therefore, degassing of the ink can be performed for a long time.

(52) When the outer support 20 is cylindrical, the outer support 20 can appropriately support the swollen hollow fiber membranes 2 from the radially outer side of the hollow fiber membrane bundle 3.

(53) When the outer support 20 has a mesh form, the swollen hollow fiber membranes 2 are supported from the radially outer side of the hollow fiber membrane bundle 3, and an increase in the pressure drop of the ink passing through the outer support 20 can be prevented. This can prevent an increase in the initial pressure of the hollow fiber degassing module 1.

(54) When the first sealing portion 8 and the second sealing portion 9 fill the spaces as described above, the plurality of hollow fiber membranes 2 divide the inner region of the housing 4 into the first region including the spaces between the plurality of hollow fiber membranes 2 and the second region including the inner spaces of the plurality of hollow fiber membranes 2. In this case, the ink supplied from the liquid supply port 6a can be supplied from the hollow portion 3b of the hollow fiber membrane bundle 3 to the spaces between the plurality of hollow fiber membranes 2 and then

discharged from the liquid discharge port **5d**. Moreover, the inner spaces of the plurality of hollow fiber membranes **2** can be evacuated through the suction port **7a**, and the ink supplied from the liquid supply port **6a** can be prevented from being discharged to the suction port **7a**.

(55) When the outer support **20** is disposed between the end surface **8b** of the first sealing portion **8** and the end surface **9b** of the second sealing portion **9**, the hollow fiber degassing module **1** can be easily produced, so that the production cost of the hollow fiber degassing module **1** can be reduced.

(56) When the end **20b** of the outer support **20** that is located on the second sealing portion **9** side is embedded in the second sealing portion **9**, the interface between the outer support **20** and the second sealing portion **9** is not exposed at the end surface **9b** of the second sealing portion **9**. In this case, even when the ink enters the interface between the outer support **20** and the second sealing portion **9**, the ink is prevented from leaking from the second sealing portion **9**. Moreover, when the suction pump **106** is connected to the suction port **7a**, breakage of the suction pump **106** due to the ink discharged from the suction port **7a** can be prevented.

(57) When the end **20a** of the outer support **20** is exposed from the first sealing portion **8**, the hollow fiber degassing module **1** can be easily produced, and the production cost of the hollow fiber degassing module **1** can be reduced.

(58) When the plurality of hollow fiber membranes **2** are not swollen and the outer support **20** is spaced apart from the hollow fiber membrane bundle **3**, the plurality of hollow fiber membranes **2** are allowed to swell, and the swollen hollow fiber membranes **2** are prevented from coming into pressure contact with the inner circumferential surface **5g** of the casing **5**. In this case, the swollen hollow fiber membranes **2** are prevented from breaking.

(59) When the outer communication portions **23** are formed in the outer support **20**, even if the outer support **20** is expanded due to, for example, the swollen hollow fiber membranes **2** to cause the outer support **20** to come into pressure contact with the inner circumferential surface **5g** of the casing **5**, the flow channel of the ink can remain between the outer support **20** and the inner circumferential surface **5g** of the casing **5**. Therefore, an abrupt increase in the pressure drop in the hollow fiber degassing module **1** can be prevented.

(60) When the inner communication portions **24** are formed in the outer support **20**, even if the outer support **20** is expanded due to, for example, the swollen hollow fiber membranes **2** to cause the outer support **20** to come into pressure contact with the inner circumferential surface **5g** of the casing **5**, the flow channel of the ink can remain between the outer support **20** and the plurality of hollow fiber membranes **2**. Therefore, an abrupt increase in the pressure drop in the hollow fiber degassing module **1** can be prevented.

(61) When the ink is as described above, the plurality of hollow fiber membranes **2** easily swell. However, since the outer support **20** is provided, an abrupt increase in the pressure drop in the hollow fiber degassing module **1** can be prevented.

(62) The preferred embodiment of the present invention has been described. However, the present invention is not limited to the above embodiment. In the embodiment described above, the inner support is provided. However, the inner support may not be provided. Even when the inner support is not provided, an abrupt increase in the pressure drop in the hollow fiber degassing module due to the plurality of hollow fiber membranes coming into pressure contact with the inner circumferential surface of the housing can be prevented. In the embodiment described above, the outer support has a cylindrical shape. However, the outer support can have any shape so long as the swollen hollow fiber membranes can be supported from the radially outer side of the hollow fiber membrane bundle. In the embodiment described above, the ink is exemplified as the liquid to be degassed. However, the liquid to be degassed may be a liquid other than the ink. Specifically, the plurality of hollow fiber membranes can swell with a liquid other than the ink. Therefore, even when such a liquid is used, the same effects as those described in the above embodiment can be obtained. In the embodiment described above, some examples of the hollow fiber degassing module have been specifically described. However, the hollow fiber degassing module can have any structure so long

as the liquid supply port and the liquid discharge port are in communication with the spaces between the plurality of hollow fiber membranes and the suction port is in communication with the inner spaces of the plurality of hollow fiber membranes so that the liquid supplied from the liquid supply port can be degassed by suction from the suction port and then discharged from the liquid discharge port. In the embodiment described above, the casing, the first lid portion, and the second lid portion included in the housing are separate members. However, they may be integrated together.

EXAMPLES

(63) Examples of the present invention will next be described. However, the present invention is not limited to the following Examples.

(64) A hollow fiber degassing module in Example 1 and a hollow fiber degassing module in Comparative Example 1 were produced. An ink was circulated using a test circuit shown in FIG. 12, and an increase in pressure drop was measured.

(65) (Test Circuit)

(66) As shown in FIG. 12, in the test circuit, a first ink supply pipe **112** inserted into an ink tank **111** storing the ink was connected to the supply port of one of the hollow fiber degassing modules **1**, and a pump **113** for sending the ink in the first ink supply pipe **112** to the hollow fiber degassing module **1** side and an inlet pressure gauge **114** for measuring the pressure of the ink in the first ink supply pipe **112** were attached to the first ink supply pipe **112**. In the test circuit, a second ink supply pipe **115** inserted into the ink tank **111** was connected to the discharge port of the one of the hollow fiber degassing modules **1**, and an outlet pressure gauge **116** for measuring the pressure of the ink in the second ink supply pipe **115** was attached to the second ink supply pipe **115**.

Example 1

(67) The hollow fiber degassing module in Example 1 was produced as follows.

(68) In Example 1, a hollow fiber degassing module SEPAREL EF-G5-B15 manufactured by DIC Corporation was used as a base module. Then, in the base module, an inner support prepared by winding a Dai-pla Netlon sheet (size of meshes: 3 mm×3 mm, thickness: 1.4 mm) manufactured by Dainippon Plastics Co., Ltd. into a cylindrical shape with a pipe diameter ϕ of 14 mm was brought into contact with the inner circumferential surface of the hollow fiber membrane bundle. Moreover, in the base module, an outer support prepared by winding a Dai-pla Netlon sheet (size of meshes: 3 mm×3 mm, thickness: 1.4 mm) manufactured by Dainippon Plastics Co., Ltd. into a cylindrical shape with a pipe diameter ϕ of 39 mm was disposed between the hollow fiber membrane bundle and the housing so as to be separated from the hollow fiber membrane bundle and from the housing. The resulting base module was used as the hollow fiber membrane module in Example 1.

(69) More specifically, hollow fiber membranes having an inner diameter of 100 μ m and an outer diameter of 180 μ m and having a side wall (membrane) made of poly-4 methylpentene-1 and having a heterogeneous structure were produced. Next, a large number of hollow fiber membranes bundled together were twisted into warp threads and woven into a bamboo blind form to thereby produce a hollow fiber membrane sheet having a prescribed length. Next, the hollow fiber membrane sheet was wound around the cylindrical inner support to thereby produce a cylindrical hollow fiber membrane bundle. Next, the hollow fiber membrane bundle and the cylindrical outer support were inserted into the casing of a housing. Then one membrane bundle end portion of the hollow fiber membrane bundle and the first end portion of the outer support were fixed to one opening end portion of the casing through the first sealing portion, and the other membrane bundle end portion of the hollow fiber membrane bundle and the second end portion of the outer support were fixed to the other opening end portion of the casing through the second sealing portion. Then the first lid portion was attached to one opening end portion of the casing, and the second lid portion was attached to the other opening end of the casing to thereby produce the hollow fiber degassing module in Example 1. The main specifications of the hollow fiber degassing module in Example 1 are shown in Table 1.

Comparative Example 1

(70) The hollow fiber degassing module in Comparative Example 1 was produced using the same procedure as in Example 1 except that the outer support was not used and that the inner support was removed from the hollow fiber membrane bundle after the resin in the first sealing portion and the second sealing portion was cured. Specifically, a hollow fiber degassing module SEPAREL EF-G5-B15 manufactured by DIC Corporation was used as the hollow fiber degassing module in Comparative Example 1. The main specifications of the hollow fiber degassing module in Comparative Example 1 are shown in Table 1.

(71) TABLE-US-00001

TABLE 1	Base module	Inner support	Outer support	Example 1	SEPAREL
Dai-pla	Netlon	Dai-pla	Netlon	EF-G5-B15	sheet manufactured by Dainippon
Size of meshes:	3 mm × 3 mm	3 mm × 3 mm	3 mm × 3 mm	Thickness: 1.4 mm	Thickness: 1.4 mm
Pipe diameter φ:	14 mm	19 mm	Compar-	SEPAREL	None
Example 1	manufactured by DIC Corporation			None	active EF-G5-B15

(Experiments)

(72) In experiments, a ceramic ink containing a hydrocarbon solvent (“Exxsol (registered trademark) D130” manufactured by Exxon Mobil Corporation (Hydrocarbones, C14-C18, n-alkanes, iso-alkanes, cyclics, aromatics, etc.)) was used as the liquid to be degassed.

(73) The temperature of the ink was set to 50° C., and the flow rate of the ink was set to 1200 g/min. Then the ink was circulated. The difference between the inlet pressure measured by the inlet pressure gauge **114** and the outlet pressure measured by the outlet pressure gauge **116** was computed as a pressure drop. The pressure drop in each of Example 1 and Comparative Example 1 after circulation of the ink for 30 minutes is shown in FIG. **13**. The relation between the time elapsed and the pressure drop during circulation of the ink for 120 hours is shown in FIG. **14**.

(74) (Evaluation)

(75) As shown in FIG. **13**, in Comparative Example 1, the pressure drop exceeded 24 kPa after the ink was circulated for 30 minutes. However, in Example 1, the pressure drop was only about 12 kPa after the ink was circulated for 30 minutes. As can be seen from these results, when the inner support and the outer support are provided, an increase in pressure drop can be reduced.

(76) As shown in FIG. **14**, in Example 1, even after circulation of the ink for 120 hours, no abrupt increase in pressure drop was found. As can be seen from these results, when the inner support and the outer support are provided, an increase in pressure drop can be prevented.

(77) In Comparative Example 1, a cross section orthogonal to the center axis and a cross section passing through the center axis after circulation of the ink for 30 minutes were observed. Moreover, in Example 1, a cross section orthogonal to the center axis and a cross section passing through the center axis after circulation of the ink for 30 minutes were observed. In Comparative Example 1, swollen hollow fiber membranes were in pressure contact with the inner circumferential surface of the casing after the ink was circulated for 30 minutes, and the swollen hollow fiber membranes entered the liquid discharge port, so that the liquid discharge port was clogged. However, in Example 1, even after circulation of the ink for one week, the outer support allowed the swollen hollow fiber membranes to be separated from the inner circumferential surface of the casing, and the swollen hollow fiber membranes did not enter the liquid discharge port. As can be seen from these results, when the outer support is provided, an abrupt increase in the pressure drop can be prevented. Moreover, as can be seen, when the thickness of the outer support is about 1.4 mm, the swollen hollow fiber membranes can be supported.

REFERENCE SIGNS LIST

(78) **1** hollow fiber degassing module, **2** hollow fiber membrane, **3** hollow fiber membrane bundle, **3a** inner circumferential surface, **3b** hollow portion, **3c** one membrane bundle end portion, **3d** the other membrane bundle end portion, **3e** outer circumferential surface, **4** housing, **5** casing, **5a** one opening end portion, **5b** the other opening end portion, **5c** side wall, **5d** liquid discharge port, **5e**

connection portion, **5f** female thread, **5g** inner circumferential surface, **6** first lid portion, **6a** liquid supply port, **6b** connection portion, **6c** female thread, **7** second lid portion, **7a** suction port, **7b** connection portion, **7c** female thread, **8** first sealing portion, **8a** communication port, **8b** end surface, **9** second sealing portion, **9a** communication hole, **9b** end surface, **10** inner support, **10a** end, **10b** end, **10c** one support end portion, **10d** the other support end portion, **11** opening, **20** outer support, **20a** end, **20b** end, **20c** one support end portion, **20d** the other support end portion, **21** opening, **21a** first opening, **21b** second opening, **22** linear portion, **22a** first linear portion, **22b** second linear portion, **23** outer communication portion, **24** inner communication portion, **101** inkjet printer, **102** ink storage unit, **103** inkjet head, **104** first ink supply pipe (ink channel), **105** second ink supply pipe (ink channel), **106** suction pump, **107** suction pipe, **111** ink tank, **112** first ink supply pipe, **113** pump, **114** inlet pressure gauge, **115** second ink supply pipe, **116** outlet pressure gauge, L axial direction, VF1 circumscribed circumferential surface, VF2 inscribed circumferential surface.

Claims

1. A hollow fiber degassing module comprising: a hollow fiber membrane bundle including a plurality of hollow fiber membranes bundled into a cylindrical shape; a housing that houses the hollow fiber membrane bundle, includes a liquid supply port and a liquid discharge port that are formed in communication with spaces between the plurality of hollow fiber membranes, and further includes a suction port formed in communication with inner spaces of the plurality of hollow fiber membranes; an outer support having a plurality of openings formed therein, supporting the plurality of hollow fiber membranes from a radially outer side of the hollow fiber membrane bundle, and disposed between the hollow fiber membrane bundle and the liquid discharge port; an inner support having a plurality of openings formed therein, supporting the plurality of hollow fiber membranes from a radially inner side of the hollow fiber membrane bundle, and disposed between the hollow fiber membrane bundle and the liquid supply port, and at least one of the plurality of openings of the outer support is formed at a position facing the liquid discharge port on the outer support, wherein the plurality of openings of the outer support include a first opening and a second opening adjacent to the first opening, and wherein the outer support has an outer communication portion that is formed between the outer support and a circumscribed circumferential surface of the outer support and allows communication between the first opening and the second opening.
2. The hollow fiber degassing module according to claim 1, wherein the outer support has a cylindrical shape.
3. The hollow fiber degassing module according to claim 1, wherein the outer support has a mesh form.
4. The hollow fiber degassing module according to claim 1, further comprising: a first sealing portion that fixes one membrane bundle end portion of the hollow fiber membrane bundle and one support end portion of the outer support to the housing; and a second sealing portion that fixes the other membrane bundle end portion of the hollow fiber membrane bundle and the other support end portion of the outer support to the housing, wherein, in a cross section passing through one membrane bundle end portion and orthogonal to an axial direction of the housing, the first sealing portion fills an entire region in the housing except for a hollow portion of the hollow fiber membrane bundle, and wherein, in a cross section passing through the other membrane bundle end portion and orthogonal to the axial direction of the housing, the second sealing portion fills an entire region in the housing except for the inner spaces of the plurality of hollow fiber membranes.
5. The hollow fiber degassing module according to claim 4, wherein the outer support is disposed between an end surface of the first sealing portion that is on a side opposite to the second sealing portion and an end surface of the second sealing portion that is on a side opposite to the first

sealing portion.

6. The hollow fiber degassing module according to claim 4, wherein an end portion of the outer support that is on a second sealing portion side is embedded in the second sealing portion.

7. The hollow fiber degassing module according to claim 4, wherein an end portion of the outer support that is on a first sealing portion side is exposed from the first sealing portion.

8. The hollow fiber degassing module according to claim 1, wherein the outer support is spaced apart from the hollow fiber membrane bundle when the plurality of hollow fiber membranes are not swollen.

9. The hollow fiber degassing module according to claim 1, wherein the outer support has an inner communication portion that is formed between the outer support and an inscribed circumferential surface of the outer support and allows communication between the first opening and the second opening.

10. The hollow fiber degassing module according to claim 1, wherein the liquid contains a hydrocarbon solvent.

11. The hollow fiber degassing module according to claim 1, wherein the liquid is at least one selected from the group consisting of glycols, glycol monoalkyl ethers, glycol dialkyl ethers, glycol monoacetates, glycol diacetates, alcohols, ketones, acetates, lactates, saturated hydrocarbons, unsaturated hydrocarbons, cyclic saturated hydrocarbons, cyclic unsaturated hydrocarbons, aromatic hydrocarbons, terpenes, ethers, cyclic imides, 3-alkyl-2-oxazolidinones, N-alkyl pyrrolidones, lactone, and nitrogen-containing solvents.

12. The hollow fiber degassing module according to claim 1, wherein the liquid is an UV ink or a ceramic ink.

13. An inkjet printer in which an ink stored in an ink storage unit is supplied to an inkjet head through an ink channel, the inkjet printer comprising the hollow fiber degassing module according to claim 1, the hollow fiber degassing module being attached to the ink channel.

14. A method for degassing a liquid, the method using the hollow fiber degassing module according to claim 1, the method comprising: supplying the liquid to the liquid supply port to thereby supply the liquid to the spaces between the plurality of hollow fiber membranes; sucking air through the suction port to degas the liquid supplied to the spaces between the plurality of hollow fiber membranes; and discharging the liquid degassed in the spaces between the plurality of hollow fiber membranes from the liquid discharge port.

15. A hollow fiber degassing module comprising: a hollow fiber membrane bundle including a plurality of hollow fiber membranes bundled into a cylindrical shape; a housing that houses the hollow fiber membrane bundle, includes a liquid supply port and a liquid discharge port that are formed in communication with spaces between the plurality of hollow fiber membranes, and further includes a suction port formed in communication with inner spaces of the plurality of hollow fiber membranes; an outer support having a plurality of openings formed therein, supporting the plurality of hollow fiber membranes from a radially outer side of the hollow fiber membrane bundle, and disposed between the hollow fiber membrane bundle and the liquid discharge port; an inner support having a plurality of openings formed therein, supporting the plurality of hollow fiber membranes from a radially inner side of the hollow fiber membrane bundle, and disposed between the hollow fiber membrane bundle and the liquid supply port, and at least one of the plurality of openings of the outer support is formed at a position facing the liquid discharge port on the outer support, wherein the plurality of openings of the outer support include a first opening and a second opening adjacent to the first opening, and wherein the outer support has an inner communication portion that is formed between the outer support and an inscribed circumferential surface of the outer support and allows communication between the first opening and the second opening.
